Project Planning Report - Group 10

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October 2026

1 Problem and Motivation



(a) Image 1, rotated (b) Image 2, flat (c) Image 3, rotated

Figure 1: Rotated image sequence

Our project task is to develop and train a network that applies object tracking to videos where the cameras orientation is changing over time. The model should be able to obtain a trajectory on a set of images similar to the ones presented in Figure 1, where image 1 and 3 are rotated, while image 2 is flat. After training, the network should identify the image rotation and edit the obtained trajectory to make it represent the objects movement in reality.

This type of network could benefit oject tracking in footage obtained from

- Handheld cameras
- Drones
- Rotating medical cameras
- Rotating CCTV cameras
- Body mounted cameras

2 Dataset

We use CDnet 2014 [1] which includes 53 videos in 11 categories with pixel-wide motion mask. We keep the static-camera categories and the masks are split into

connected components. We focus on data with only one moving object, so we filter out categories with non-static cameras and multiple moving objects. Data is split by video so we can get a valid test/training split.

3 Available Code & Papers

For object detection, there are multiple pretrained models available online, for example YOLO from Ultralytics. These can be used in the torch.nn format, which overlaps with earlier course knowledge. Torch.nn can also perform image augmentations. For evaluation, there is a python library called trackeval, which allows for multiple different metrics to be evaulated.

The following papers are found to be relevant to the project:

- Background subtraction for the moving camera: A geometric approach [2]
- Real-time tracking with stabilized frame [3]
- Statistical background modeling for non-stationary camera [4]

4 Evaluation

We plan to evaluate the model by comparing its predicted rotation angle with the image's starting orientation. After correcting the angle rotation, we will evaluate how accurately the model estimates an object's trajectory between consecutive frames. This by comparing predicted displacement of an object, with the displacement of the object given by the annotated data.

The prediction errors will be calculated to measure the model's performace. We will also test different variations of the model through an ablation study. This will allow us to explore how correcting the rotation of a frame affects the accuracy of the estimated movement of an object.

5 Time Plan

- **SW6:** Planning report, data download and sorting.
- **SW7:** Format labels, setup and train model. Evaluation and ablations
- **SW8:** Analysis, poster and report creation. Poster due Thursday 23:59.
- **SW9:** Final report due Monday 23:59. Project presentations.

References

[1] Y. Wang *et al.*, "CDnet 2014: An expanded change detection benchmark dataset," in *Proc. CVPR Workshops*, 2014.

[2] D. Zamalieva and A. Yilmaz, "Background subtraction for the moving camera: A geometric approach," *Comput. Vis. Image Underst.*, vol. 127, pp. 73–85, 2014, doi: 10.1016/j.cviu.2014.06.007.

[3] Z. Wang, Z. Zhao, and F. Su, "Real-time tracking with stabilized frame," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recogn. Workshops (CVPRW)*, 2020.

[4] Ying Ren, Chin-Seng Chua *, Yeong-Khing Ho, "Statistical background modeling for non-stationary camera" in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recogn. Workshops (CVPRW)*, 2001.